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The equivalent circuit method

Quick Guide to Using *ECM_MS* Software for 3D Magnetostatics: Examples

https://github.com/JNSresearcher/ECM_MS

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Example 1.

Here are some guidelines for constructing a magnetostatic calculation problem.

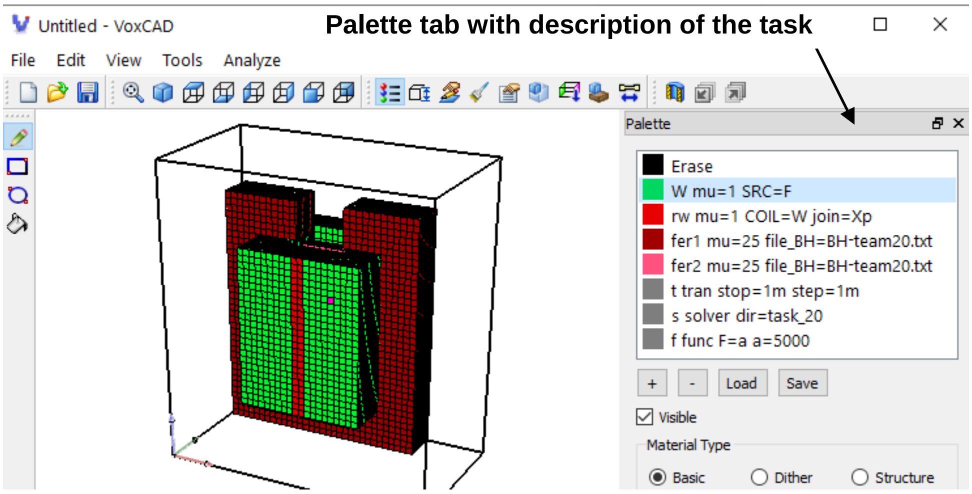


Fig.1. General view of the editor ***VoxCAD***.
On the left is the workspace; on the right is the palette tab.

The text data of the task is entered in the **Palette** tab.

General notes. First, the strings defining the geometry and properties of the domains are placed. Domains are displayed in the workspace. Then come the lines containing the modeling information. (various functions, setting time intervals, etc.). The line begins with a name. Except for lines describing coils, this name is not used. In strings containing modeling information, the line name must be followed by a keyword defining the purpose of the line (e.g. solver, function, etc.) and then sequence of keyword-value pairs. These strings should not be displayed in the work area.

The case of the characters does not matter, as the characters are converted to uppercase during processing. In our case we have the following text:

```
W mu=1 SRC=F
rw mu=1 COIL=W join=Xp
fer1 mu=25 file_BH=BH-team20.txt
fer2 mu=25 file_BH=BH-team20.txt
t tran stop=1m step=1m
s solver dir=task_20
f1 func F=a a=5000
```

* **W** string. This is where the coil information is located. The first word is the coil name, **W**. Next, the keyword **mu** is the relative magnetic permeability. Since the coils must always be located in an air environment, then **mu=1**. Then the keyword **SRC** and a reference to a function named **F**. Function **F** determine the amplitude of the current in the coil.
* **rw** string. The main purpose of this string is to indicate the direction of current in the coil. The domain must consist of two cells and be located in the coil's cross-section, in a plane perpendicular to the current direction. In this case, the cross-section is in the **YZ** plane (perpendicular to the **X** axis). The string must contain the keyword **COIL** indicating the name of the coil and the keyword **JOIN** indicating the direction of the current. In this example, a reference is made to a coil named **W**, and the direction of current in this section is along the positive direction of the **X** axis, and is designated as **Xp** (for the opposite direction, **Xm** must be specified). If the sections are located in other planes, then the designations **Yp**, **Ym**, **Zp**, **Zm** are acceptable for the directions (see section *2*).
* **fer1** string. **mu=25**. Information about the relative magnetic permeability of the core. In this case **μ = 25** - initial approximation of magnetic permeability. It is recommended to select initial values in the range of 5-25. The next keyword in this line: **file_BH** indicates that the region has nonlinear magnetic permeability, and the magnetization curve is represented by table **B-H**. In this example, the data is contained in the file **BH-team20.txt**. Example file contents:

```
0.0      0.0
0.01     27
0.025    58
0.05     100
  .....
2.3    135000
2.31   140000
2.33   150000
```

If the **file_BM** keyword is specified, the file must contain **B-μ** values.
* **fer2** string. Identical to the **fer1** row, that is, the second ferromagnet has the same properties.
* **t** string. After the keyword **tran -** the parameters of the transient process in seconds are set:

  The keyword **stop** is the time of the transient process. This is a variant of the value with a prefix: ***1m***, i.e. *1e-3 sec*. (More about *prefixes*);

  The keyword **step -** is the time step;

  You can use the keyword **jump -** is the time jump for outputting the results to a file.

  In this example, one time step is sufficient for the magnetostatic problem.
* **s** string. After the word **solver** - the solver parameters are set:

  The keyword **dir** - specifies the name for the output directory, in this example it is **`task_20`**.

The following keywords are possible (default values in brackets):

TOLSOR (0.5m) – convergence criterion for SOR

ITSOR (10000) – maximum number of iteration for SOR

OMESOR (1.92) – over-relaxation coefficient for SOR

TOLMAG (1m) – convergence criterion for the nonlinear part of the equations

ITMAG (200) – number of iteration for the nonlinear part of the equations

OMEMAG (0.0) – under-relaxation coefficient for nonlinear equations (from the range 0-1, best value = 0.7)

* **f1** string. After the word **func**, the description of the function **F** is entered: after the **"="** sign, without spaces, the symbolic expression of the function is written, then the names and values of the arguments. For example, the symbolic expression: **a*cosd(360*f*t)** has 3 arguments **a**, **f, t**. Numeric values are assigned to the arguments immediately after the function. The **a** argument specifies the amplitude of the current in the coil (*18 AT*). The **f** argument set the frequency of the current in the coil. The keyword **t** is assigned to the argument **t** "t". During the calculation, the calculated time will be assigned to this argument. In this example, the function is a simple constant **a**, which is assigned the value *5000.* That is, in the example, the coil is assigned a current of 5000 ampere-turns

Note:

Function arguments may contain algebraic expressions enclosed in quotation marks. These expressions may include numbers (without prefixes) and symbolic constants. The following symbols may be used to denote constants:

dx, **dy**, **dz** - grid steps along **x,y,z**.

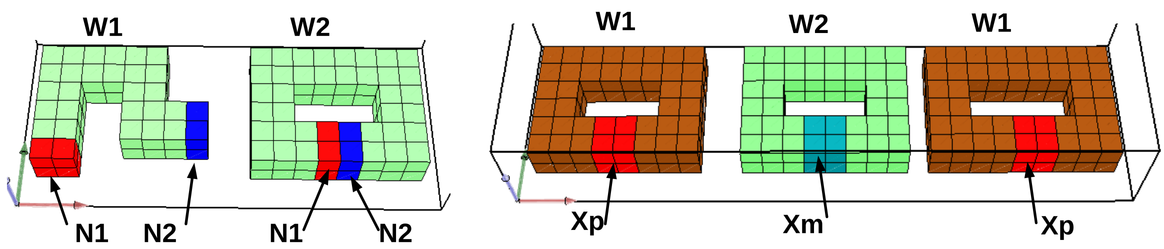
Nx, **Ny**, **Nz** - size of the calculation area along **x,y,z**.

time, **dt, t** - transient time, time step and current time.

pi, e, **mu0**, **e0** : $\pi = 3.1415...$, $e=2.7182...$, $\mu_0=0.12566...10^{-5}$, $\varepsilon_0=0.88541...10^{-11}$, respectively.

2 Creation of direction of external current sources in conductors

In the cross-section of the conductor, it is necessary to place “contacts”. There are two methods: in the first, a contact is placed at the beginning (designated by the keywords **join=N1**) and end (designated by the keywords **join=N2**) of the conductor. Each such contact must have a thickness of *1* cell. In the second method, intended only for closed conductors (coils), only one contact, two cells thick, is placed in the coil's cross-section. The direction of the current is specified along the coordinate axes using the keywords **join=** and one of the following: **Xm, Xp, Ym, Yp, Zm, Zp**. If the coils are identical in shape, they can be connected in parallel. After the keyword **par**, the number of parallel coils is indicated.



a) Setting the start (**N1**) and end (**N2**) of current-carrying conductors

b) A double-layer contact in the coil cross-section
Coil **W1** consists of two parts connected in parallel.

Fig.2. Creating a direction of current in conductors

Below are parts of the tasks for Fig. 2. A constant current of *10 A* is set in all conductors.

<u>part of the task for Fig.2 a)</u>:

```
W1 mu=1 SRC=F
r11 mu=1 COIL=W1 JOIN=N1
r12 mu=1 COIL=W1 JOIN=N2
W2 mu=1 SRC=F
r21 mu=1 COIL=W2 JOIN=N1
r22 mu=1 COIL=W2 JOIN=N2
……
f func F=a a=10
```

<u>part of the task for Fig.2 b):</u>

```
W1 mu=1 SRC=F par=2
r1 mu=1 COIL=W1 JOIN=Xp
W2 mu=1 SRC=F
r2 mu=1 COIL=W2 JOIN=Xm
……
f func F=a a=10
```

Note. When drawing a coil, aim for a uniform cross-section along its entire length. If the coil's cross-section is too small (usually at “bends”), this may result in warning messages. The explanation is given in Fig.3.

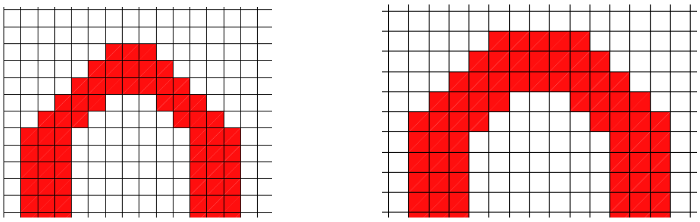


Fig.3. On the left, the coil width at “the bends” is reduced (to 2 cells).
On the right, the coil width at all “bends” is 3 cells.

3 Example 2. Linear synchronous machine (LSM).

A three-phase winding is placed in a stationary magnetic core and creates a moving magnetic field, first in the positive **x**-axis direction, then in the opposite direction. The excitation winding is located on a ferromagnetic core and is supplied with direct current. The winding and core move synchronously with the field of the three-phase winding. The velocity of the field **Vs** of the three-phase winding is $2 \cdot \tau \cdot f$, where τ is the pole pitch, **f** is the frequency of the three-phase winding current. Three coils of a stationary magnetic core are shifted in space by 120 degrees, and a current flow through them that is shifted in time by 120 degrees.

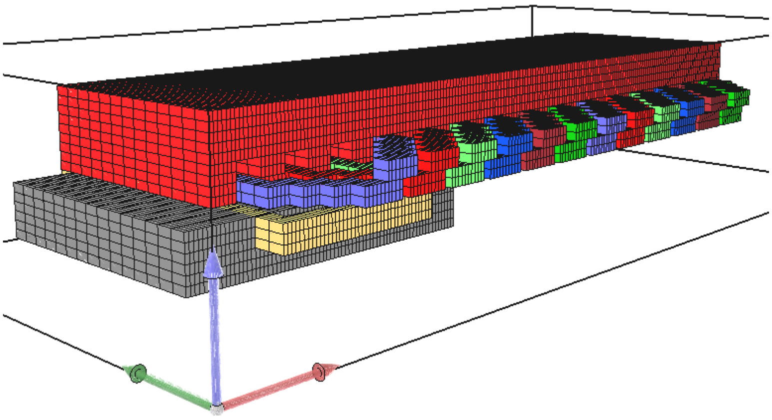


Fig.4. Screenshot of **LSM** in ***VoxCAD***

Data for calculation:

```
A mu=1 SRC=Ia par=2
B mu=1 SRC=Ib par=2
C mu=1 SRC=Ic par=2
rA mu=1 COIL=A JOIN=Yp
rB mu=1 COIL=B JOIN=Yp
rC mu=1 COIL=C JOIN=Yp
Fe mu=100
Fe2 mu=100 Vex=Vx
A2 mu=1 SRC=Ia par=2
rA2 mu=1 COIL=A2 JOIN=Yp
B2 mu=1 SRC=Ib par=2
rB2 mu=1 COIL=B2 JOIN=Yp
C2 mu=1 SRC=Ic par=2
rC2 mu=1 COIL=C2 JOIN=Yp
F0 mu=1 SRC=Fe Vsx=Vx
rF0 mu=1 COIL=F0 JOIN=Yp
p1 tran stop=100m step=1m
p2 solver dir=LSM tolsor=1m
f1 func Ia=a*cosd(360*fs*t) a=10 fs=28.6 t=t
```

```
f2 func Ib=a*cosd(360*fs*t-impl2(sind(360*fm*t))*120) a=10 fs=28.61 fm='1/time' t=t
f3 func Ic=a*cosd(360*fs*t+impl2(sind(360*fm*t))*120) a=10 fs=28.61 fm='1/time' t=t
f4 func Fe=a a=2.5
f5 func Vx=2*tau*fs*impl2(sind(360*fm*t)) tau='36*dx' fs=28.61 fm='1/time' t=t
```

A, **B**, and **C** strings correspond to regions of the three-phase winding and contain a reference to the name of the function that determines the current in the coil (after the keyword **SRC**), as well as an indication of the parallel connection of the two coils of each phase (after the keyword **par**). The string begins with the name of the coil. The coils must be located in air, so **μ** is always equal to *1*.

rA, **rB**, **rC** strings correspond to the cells located in the cross-section of the corresponding coils and contain a reference to the coil name (after the keyword **COIL**) and an indication of the positive direction of the current relative to the **Y**-axis (after the keyword **join**).

Fe string corresponds to stationary magnetic core **μ=100**

Fe2 string corresponds to moving magnetic core, **μ=100**. The velocity of movement in the **x**-axis direction is determined by the **Vex** keyword, followed by the name of the function describing the movement; in this example, the function name is **Vx**. The **Vey** and **Vez** keywords are also possible for other directions.

NOTE For moving ferromagnets, nonlinearity is not provided.

A2, **B2**, and **C2** strings - same as **A**, **B**, and **C** strings

rA2, **rB2**, **rC2** strings - same as **rA**, **rB**, **rC** strings

F0 string corresponds to region with excitation coil. This coil moves with the ferromagnetic core. The velocity of movement in the **x**-axis direction is determined by the keyword **Vsx**, followed by the name of the function describing the motion; in this example, the function name is **Vx**.

rF0 string contain a reference to the coil name **F0**

The following strings do not apply to domains:

p1 string. After the keyword **tran -** the parameters of the transient process in seconds are set:

The keyword **stop** is the time of the transient process. This is a variant of the value with a prefix: *100m*, i.e. *100e-3 sec*. (More about *prefixes*). The keyword **step -** is the time step (*1m*), that is, there will be an output of *100* points. You can use the keyword **jump** - is the time jump for outputting the results to a file, For example, **jump=2m**, the results will be displayed every *2msec*, that is, we will get not *100* points but *50* points;

p2 string. After the word **solver -** the solver parameters are set:

The **dir** keyword specifies the name of the output directory: **LSM**.

The **tolsor** keyword specifies convergence criterion for SOR In this example it is *1m*

f1, **f2**, and **f3** strings describe functions of time that correspond to the currents in the coils. These are cosine curves, shifted in time by *120* degrees, with amplitude of *10AT* and a frequency of *28.6 Hz*. In addition, the **impl2** function changes the sign of the phase shift of **B** and **C** to change the direction of field motion.

f4 string contains a function that sets the current in the excitation winding

f5 string contains a function that sets the speed of movement of the excitation winding. This function, **impl2**, converts a sine wave into rectangular pulses with positive and negative amplitudes ***tau*fs***. This means that the plate initially moves at a constant positive velocity, and then, after half the specified transient time, the velocity changes to a constant negative velocity. ***fs*** is current frequency in the coils, ***tau*** is the pole pitch.